

CS 1925 Calculus II Final Exercise

Name: _____

St. ID: _____

**10 points to each question

1. Finding the following Partial Derivatives

(a) $\frac{\partial}{\partial z}[xy + xz]$ x

(b) $\frac{\partial}{\partial z}[z \sin(xz + 2z)]$ $\sin(xz + 2z) + z(x+2)\cos(xz + 2z)$

2. Find the partial Derivatives for $f(x, y) = xy e^{xy}$

(a) $f_x(1, \ln 5)$ $5 \ln 5 (1 + \ln 5)$

(b) $f_y(2, \ln 3)$ $18(1 + 2 \ln 3)$

3. $f(x, y, z) = ye^x \sin z + xy \ln z$, find

(a) f_{xz} $ye^x \cos z + \frac{y}{z}$

(b) f_{zz} $-ye^x \sin z - \frac{xy}{z^2}$

4. Evaluate the following iterated integrals

(a) $\int_0^2 \int_0^2 x \sin y dx dy$ $2(1 - \cos 2)$

(b) $\int_0^2 \int_0^{\sqrt{4-y^2}} \frac{2}{\sqrt{4-y^2}} dx dy$ 4

5. Evaluate the following iterated integrals

(a) $\int_0^\pi \int_0^{\cos \theta} r dr d\theta$ $\frac{\pi}{4}$

(b) $\int_0^{\frac{\pi}{2}} \int_0^1 (r \cos \theta)(r \sin \theta) r dr d\theta$ $\frac{1}{8}$

6. Find the area of the region bounded by $r = 3 \cos 3\theta$, the area of the curve traced as θ increases from $-\pi/6$ to $\pi/6$. $\frac{3}{4}\pi$

7. Solve the following Differential Equations

(a) $\frac{dy}{dx} = y + 1$ $y = Ke^x - 1$

(b) $xyy' = 3$ $y = \pm \sqrt{6 \ln|x| + C}$

8. Find the following indefinite integrals

(a) $\int x^3 \ln x dx$ $\frac{1}{4}x^4 \ln x - \frac{1}{16}x^4 + C$

(b) $\int \frac{x e^{2x}}{(2x+1)^2} dx = \frac{e^{2x}}{4(2x+1)} + C$

9. Find the following indefinite integrals

(a) $\int x \cdot \cos x dx$ $x \sin x + \cos x + C$

(b) $\int \sin x \cdot e^x dx$ $\frac{e^x}{2}(\sin x - \cos x) + C$

10. Find the following indefinite integral

$\int x \cdot \sin x \cdot e^x dx$ $\frac{e^x}{2}(x \sin x - x \cos x + \cos x) + C$

Good Luck!